

Optimal Film Portfolio Allocation Under Box Office Uncertainty

1. Introduction

For many years, **Whispering Pencil Studios**, a US horror and animation production company, has produced films based on a variable annual budget and the greenlight proposals received that year. However, after many years of variable success, we decided to take a more structured approach to optimize our revenue.

After much deliberation, we turned to the data. We filtered a Kaggle dataset of over 45,000 movies to only include movies in English, from the United States, and in either the Horror or Animation genre. Horror and Animation were chosen due the high ROI. After cleaning the data, we utilize 2-Stage Stochastic Optimization, which is discussed in detail in Section 2. Using this framework instead of a one-stage framework allows the team to account for several factors of real-world uncertainty. As a result, we believe that we can improve our total revenue over time by using our model results to focus on making films based on specific quotas and specific budget guidelines.

2. Problem Formulation

2.1 Uncertainties

The primary uncertainty is per-film box office revenue, R_{ij} , for genre i and budget tier j . Using @RISK, we fit revenue distributions to 804 decade-weighted (Table 1) historical films across six genre-tier segments shown in Table 2:

DECADE WEIGHTS (Recency Bias)	
1915–1979	0.2
1980–1989	0.4
1990–1999	0.6
2000–2009	0.8
2010+	1.0

Table 1. Decade weighting for revenue

Genre	Budget Tier	Min Budget	Max Budget	Distribution	Mean Weighted Revenue	N
Horror	Low	\$12000	\$5M	LogNormal	\$17.5M	210
Horror	Medium	\$5M	\$19M	LogNormal	\$34.2M	185
Horror	High	\$19M	\$200M	LogNormal	\$77.4M	159
Animation	Low	\$590000	\$51M	LogNormal	\$48.4M	104
Animation	Medium	\$51M	\$95M	LogNormal	\$259.7M	76
Animation	High	\$95M	\$260M	LogNormal	\$418.3M	70

Table 2. Characteristics of genre-budget tier segments

Distributions were fitted using AIC-minimizing criteria. To additionally represent their right-skewed distributions, LogNormal distributions were used. Decade recency weights (0.2–1.0) reduces the influence of older films to reflect current market conditions.

Other uncertainties are estimated: budget is estimated by the average of each budget tier. Audience cannibalization in Stage 2 is estimated by each film beyond a threshold of 5 films per segment within a release year incurs a 5% revenue penalty on the excess films to reflect market saturation. The CvaR risk penalty in Stage 2 is estimated by $\lambda \times \text{CVaR}_{95}$, where CVaR is approximated as $1.65 \times$ portfolio standard deviation and $\lambda = 0.30$ (moderate risk aversion). It represents the volatility of the performances of the bottom 5%.

2.2 Upfront Decision Variables

In Stage 1, N_{ij} , the number of films to greenlight for genre $i \in \{\text{Horror, Animation}\}$ and budget tier $j \in \{\text{Low, Medium, High}\}$, is used to set the **budget** for the next year, B_{S1} and determine N_{total} , the max total number of films.

$$N_{ij} \in \mathbb{Z}_{\geq 0} \text{ for } i \in \{\text{Horror, Animation}\}, j \in \{\text{Low, Medium, High}\}$$

2.3 Wait-and-See Decision Variables

In Stage 2, after committing to a budget and max total number of films, the studio refines its allocation via x_{ij} , the final film count per segment, by also considering audience cannibalization and CvaR risk penalty.

2.4 Objective Function

Stage 1 maximizes the 50th percentile of the net revenue due to lognormal distributions:

$$\max \sum_{ij} N_{ij} \cdot R_{ij} - \sum_{ij} N_{ij} \cdot B_{ij}$$

Stage 2 maximizes the 50th percentile of the risk-adjusted net revenue:

$$\max \sum_{ij} x_{ij} \cdot R_{ij} - \sum_{ij} \text{Cannibalization}(x_{ij}) - \sum_{ij} x_{ij} \cdot B_{ij} - \lambda \cdot \text{CVaR}_{95}$$

2.5 Constraints

- Budget Stage 1: $\sum_{ij} N_{ij} \cdot B_{ij} \leq \300M
- Budget Stage 2: $\sum_{ij} N_{ij} \cdot B_{ij} \leq B_{S1}$
- Genre proportion Stage 1: Horror share $\leq 68.85\%$ of total films (mirrors historical dataset split)
- Genre proportion Stage 2: Animation Share $> 0\%$ of total films
- Number of Films Stage 2: $\sum_{ij} x_{ij} \leq N_{\text{total}}$
- Integer film counts: $N_{ij}, x_{ij} \in \mathbb{Z}_{\geq 0}$ - cannot greenlight a fraction of a film
- Non-negativity: $N_{ij}, x_{ij} \geq 0$

3. Solution and Recommendation

3.1 Solution

Stage 1 — Total Production Budget

Stage 1 maximizes expected net revenue, which is calculated by total portfolio revenue minus total production cost. It is subjected to a budget constraint of \$300M. Using @Risk Optimizer on Microsoft Excel, the optimal solution is to allocate \$299.8M in total budget, which could fund 16 films and generate \$907.7M in revenue, with a net profit of \$607.9M in the ideal simulation. However, as Risk Optimizer involves simulations, these revenue values would be better represented by a range. The 50th percentile was \$528.3M.

Genre / Tier	# Films	Avg Budget	Budget	Revenue/Film	Total Revenue	Net Profit
Horror — Low	10	\$2.4M	\$24.0M	\$8.2M	\$82.2M	\$58.2M

Horror — Medium	1	\$11.6M	\$11.6M	\$14.0M	\$14.0M	\$2.4M
Horror — High	0	\$42.4M	—	\$77.4M	—	—
Animation — Low	2	\$21.4M	\$42.8M	\$22.7M	\$45.3M	\$2.5M
Animation — Medium	3	\$73.8M	\$221.4M	\$255.4M	\$766.2M	\$544.8M
Animation — High	0	\$144.6M	—	\$417.5M	—	—
TOTAL	16	—	\$299.8M	—	\$907.7M	\$607.9M

Table 3. Stage 1 Film Allocation by Genre and Budget Tier, Run 1

The solution is driven by two genres and tiers with the highest ROI, Horror Low and Animation Medium. The resulting split of 68.75% Horror / 31.25% Animation mirrors the dataset split of 68.85% / 31.15%.

Stage 2 — Film Count and Tier Allocation

Stage 2 refines the portfolio allocation by explicitly incorporating two real-world risks absent from Stage 1: audience cannibalization and CVaR risk penalty. It is also constrained by the budget and total number of films set in Stage 1. The optimal solution is to invest in 12 films. In one scenario, we see that it can produce a net revenue of \$325.5M. In this model, the 50% percentile is \$362.6M.

Genre / Tier	# Films	Avg Budget	Cannibalization	Revenue
Horror — Low	5	\$2.4M	—	\$76M
Horror — Medium	4	\$11.6M	—	\$54.8M
Horror — High	0	\$42.4M	—	—
Animation — Low	1	\$21.4M	—	\$102.2M
Animation — Medium	1	\$73.8M	—	\$118M
Animation — High	1	\$144.6M	—	\$468.3M
TOTAL	12	—	—	\$819.5M

Table 4. Stage 2 Film Allocation by Genre and Budget Tier

Stage 2 results have several differences from Stage 1. The total approved film count falls from 16 to 12 because Stage 2's risk penalty and cannibalization costs make marginal films less attractive, particularly when making more than 5 Horror Low films, since it would trigger an additional 5% revenue reduction across all Horror Low films in the portfolio. Therefore, Horror Low is reduced from 10 to 5 films.

Stage 2 also introduces one Animation High film. At this portfolio size, the budget freed by reducing Horror Low exposure is sufficient to accommodate Animation High's large cost while keeping

total spending at \$298.8M. The final genre split - 75% Horror / 25% Animation – highlights focusing on horror films but including riskier animation films as well.

3.2 Recommendation

Based on the two-stage optimization, we offer the following actionable recommendations for Whispering Pencil Studios:

- **Adopt the Stage 2 portfolio as the annual greenlight template.** Produce 12 films per year: 5 Horror Low, 4 Horror Medium, 1 Animation Low, 1 Animation Medium, and 1 Animation High. This allocation respects the \$300M budget (\$299.8M used), and maximizes risk-adjusted net profit.
- **Prioritize Animation Medium and High despite high upfront cost.** The revenue distribution for Animation Medium and High means rare blockbusters dominates returns; even one outsized hit justifies the allocation.
- **Cap Horror Low at 5 films annually.** Beyond this threshold, audience cannibalization erodes returns by 5% per additional film. Additional Horror budget is better redeployed into Animation tiers where cannibalization does not apply.
- **Use Horror Low as a portfolio anchor, not a growth engine.** Horror Low films are high in count but can be individually low in revenue. Their value lies in diversification, where many small, inexpensive bets smooth out downside variance, and not in generating outsized returns. Whispering Pencil should continue funding these films as portfolio stabilizers while directing growth investment toward Horror Medium and Animation.
- **Monitor CVaR annually and recalibrate λ .** The 0.30 risk aversion weight is a calibration choice. In years with tighter capital constraints or adverse market conditions, increasing λ will shift the optimizer toward safer, lower-variance allocations. Embedding CVaR as a dynamic, annually-reviewed parameter rather than a fixed constant will improve long-run portfolio resilience.
- **Wait-and-see: scale Animation High marketing contingently.** Given Animation High's large budget and high revenue variance (\$268M std dev), commit full P&A spend only after strong early-weekend performance signals. A revenue threshold of $\geq 15\%$ of production budget in opening weekend is a reasonable trigger for full international marketing expansion.

4. Discussion and Future Work

4.1 Discussion

The two-stage stochastic optimization model developed in this project offers a structured framework for film portfolio planning under uncertainty; however, it comes with certain limitations that warrant reflection. The main challenge lies in the nature of the film industry itself as revenue outcomes are driven by a large range of factors: including timing, viewer reception, marketing execution, involved talent, and competing releases. many of which complicates quantification. By reducing this complexity to a distribution by genre and budget tier, we capture the broad shape of uncertainty but lack additional factors that make any individual film succeed or fail in the box office.

The central trade-off in any optimization mode is complexity versus tractability. To keep the model solvable and interpretable, we had to make certain assumptions, such as the historical weighted averages and standard deviations are suitable approximations for future revenue distributions, and using a Monte Carlo simulation of those distributions to represent real world stochasticity. In reality, film revenues are distributions that could be difficult to capture, even with a LogNormal fitting. The weighting by decade addresses the issue of data relevance by downweighting older films, but the dataset still reflects market conditions, audience sizes, streaming disruption, theatrical windows, that have shifted substantially. This is a limitation that no weighting scheme can fully resolve without richer, more granular data.

Also, modern machine learning methods present compelling opportunities to improve revenue forecasting. Gradient boosting models trained on features such as director track record, franchise affiliation, release timing, and marketing spend could yield substantially better point predictions than tier-

level averages. This method could be combined with stochastic optimization, effectively coupling data-driven forecasting with decision optimization - a more realistic and robust approach to the problem.

4.2 Future Work

With more time, we would prioritize the following improvements:

1) Dynamic CVaR Integration

A meaningful improvement would embed CVaR dynamically into the objective function - Maximize: $E[\text{Revenue}] - \lambda \times E[\text{Loss} \mid \text{worst 5\% scenarios}]$ - so that every simulation contributes to the optimization. This formula will also force the solver to find allocations that are robust across the full distribution rather than optimized only for favorable scenarios.

2) Addressing the 23.5% Feasibility Rate

Only 235 of 1,000 Monte Carlo trials converged to a feasible solution, largely because the Horror proportion constraint ($\leq 68.85\%$), which was derived from the historical dataset composition rather than market demand data, could not be satisfied in most random revenue scenarios. The optimal value of \$1.158B should therefore be interpreted as a conditional expectation across favorable scenarios only and not an unconditional expected return. Future work should empirically validate this constraint and consider replacing it with a soft penalty to avoid widespread infeasibility.

3) Budget Uncertainty

Production costs are treated as fixed within each tier. In practice, film budgets routinely overrun by 20–50%. Modeling the budget as a second stochastic variable by using a right-skewed distribution around the tier average would yield more conservative and realistic net revenue estimates. Furthermore, it would interact with the CVaR mechanism to more accurately capture downside scenarios.

4) Expanded Genre Coverage and Cross-Genre Correlations

The current model covers only Horror and Animation. Extending to Action, Comedy, Drama, and Sci-Fi would allow the optimizer to exploit cross-genre diversification benefits, particularly where revenue correlations between genres are low or negative within the same release window. Cross-genre correlation is where diversification value is most practically meaningful for a real studio portfolio.

Signature

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